N MOLECULAR / V3—V11

From structural graph probes to executable molecular computations

A source-grounded research portfolio, figures, quantitative receipts and reproducibility record

Research archive snapshot: 23 October 2026. Public scientific references checked on the same date. All figures are generated from recovered on-disk experimental records—not illustrative source images.

RECOVERED EVIDENCE	CONFIRMED VALUE
Computational paired comparisons (V9–V11)	13,312
V11 source-built paired comparison records	7,936
V11 actual reset SNN executions	15,872
Historical V6–V8 graph-classifier result rows	100
V11 named passing Python tests	81 (historical log)
Independent disagreement arithmetic errors	0 of 13,312
V11 recorded pipeline wall time	32.067 seconds
V11 molecular MMFF94 minimizations	13 converged

Research claim. A deterministic native spiking neural system responds to graph interventions derived from known sequences, computed secondary structures, and generated atomic geometries. Sensitivity is verified; experimental three-dimensional folding accuracy and ligand activity prediction have not been demonstrated.

Reproducibility contract. Each exhibit names its evidence table. The bundle contains the complete recovered raw audit evidence and an independent verifier; the large native C/C++ project is supplied separately as the archived source release.

01 / WHAT WAS ACTUALLY EXECUTED

Research chronology and evidence hierarchy

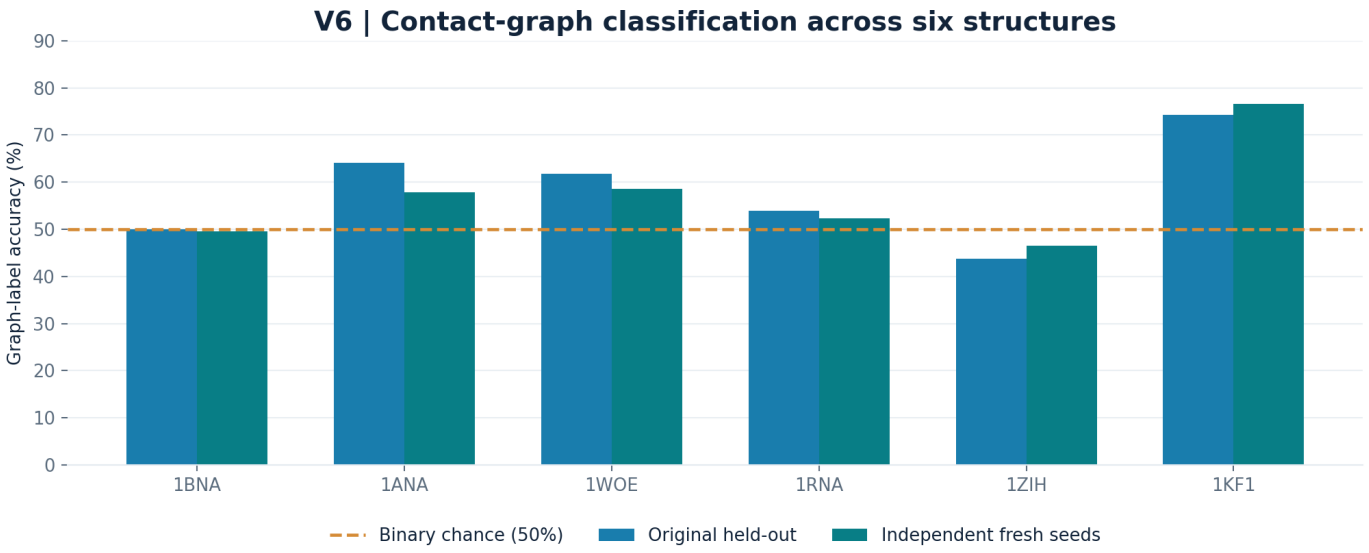
STAGE	EXPERIMENTAL UNIT	WHAT WAS REPORTED
V3–V5	Native molecular input and SNN trials	Source bridge, original 6-case series and controls
V6	128 additional deterministic excitation seeds	Fresh graph classifier tests; 1KF1 motif-preserving falsification
V7–V8	Four additional RNA structure labels	8,192 new recorded trial rows; full replay audits
V9	Eight chemical SMILES graphs	2,048 pairs / 4,096 resets / 48 ticks
V10	5 RNA + 8 ligand inputs	3,328 pairs / 6,656 resets / 48 ticks
V11	Thermo, PK candidate, geometry & force branches	7,936 pairs / 15,872 resets / 48 ticks

Evidence classes. Original experiment output (JSONL) is the strongest record for firing behavior; the source-driven audit CSVs reconstruct those outputs. Scientific reports and run logs document statistical protocols, tests and timing. Experimental PDB metadata identify molecules but do not automatically validate the computed structure.

Not additive biological sample counts. The V8 historical 22,528-trial set describes earlier deterministic networks; V9–V11 trials are different interventions, often sharing molecules and methodological ancestors.

EXHIBIT 01 / SOURCE-RENDERED DATA

Native graph-label discrimination: original versus fresh seeds

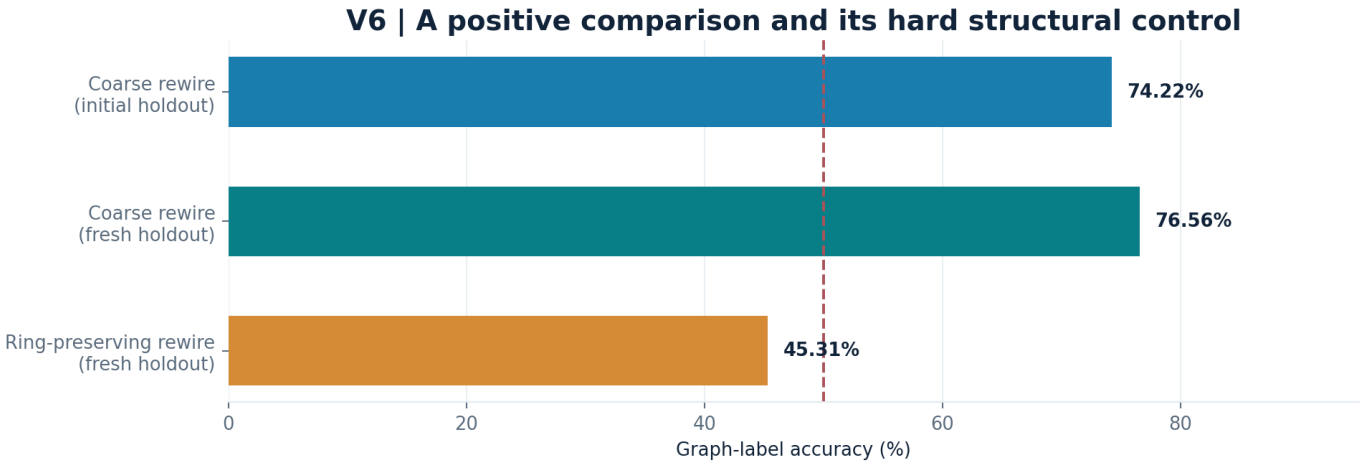


Source: GRAPH_CLASSIFICATION_CONDITIONS.csv

Across six hand-authored DNA/RNA contact-graph contrasts, some raw neural readouts carried decodable intervention information on previously unused stimulation seeds. V6 values classify a computational rewire label, not a physical fold.

EXHIBIT 02 / SOURCE-RENDERED DATA

One decisive negative control changes the interpretation

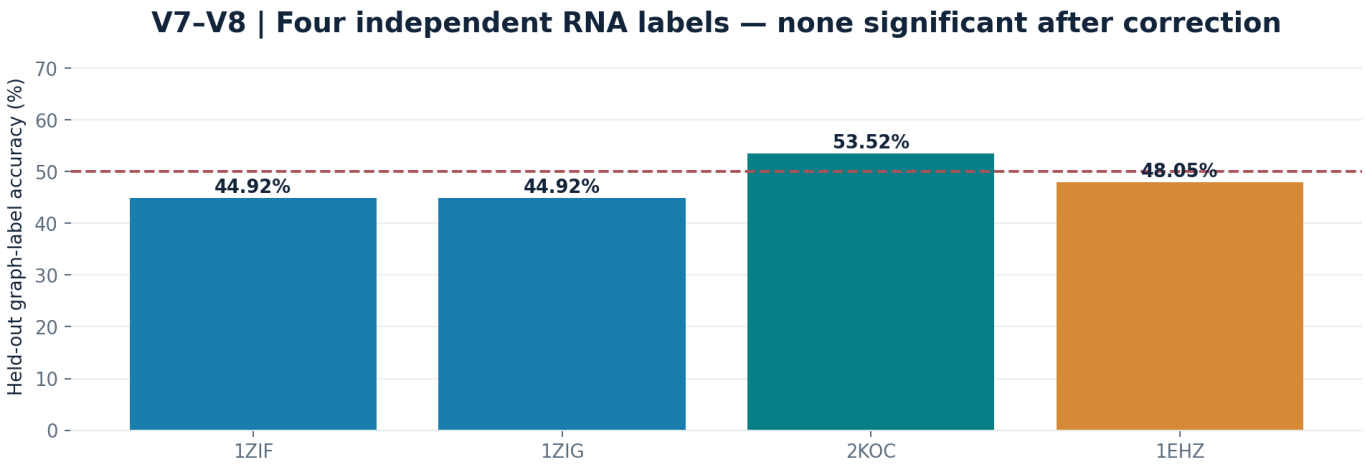


Source: Graph classifier CSV and V6 scientific report

For 1KF1, the fresh cycle-breaking comparison reached 196/256 correct (76.56%); preserving four-site rings yielded 116/256 (45.31%), close to chance. This is compatible with coarse cycle sensitivity, not generalized molecular geometry inference.

EXHIBIT 03 / SOURCE-RENDERED DATA

Additional RNA cases: corrected null results

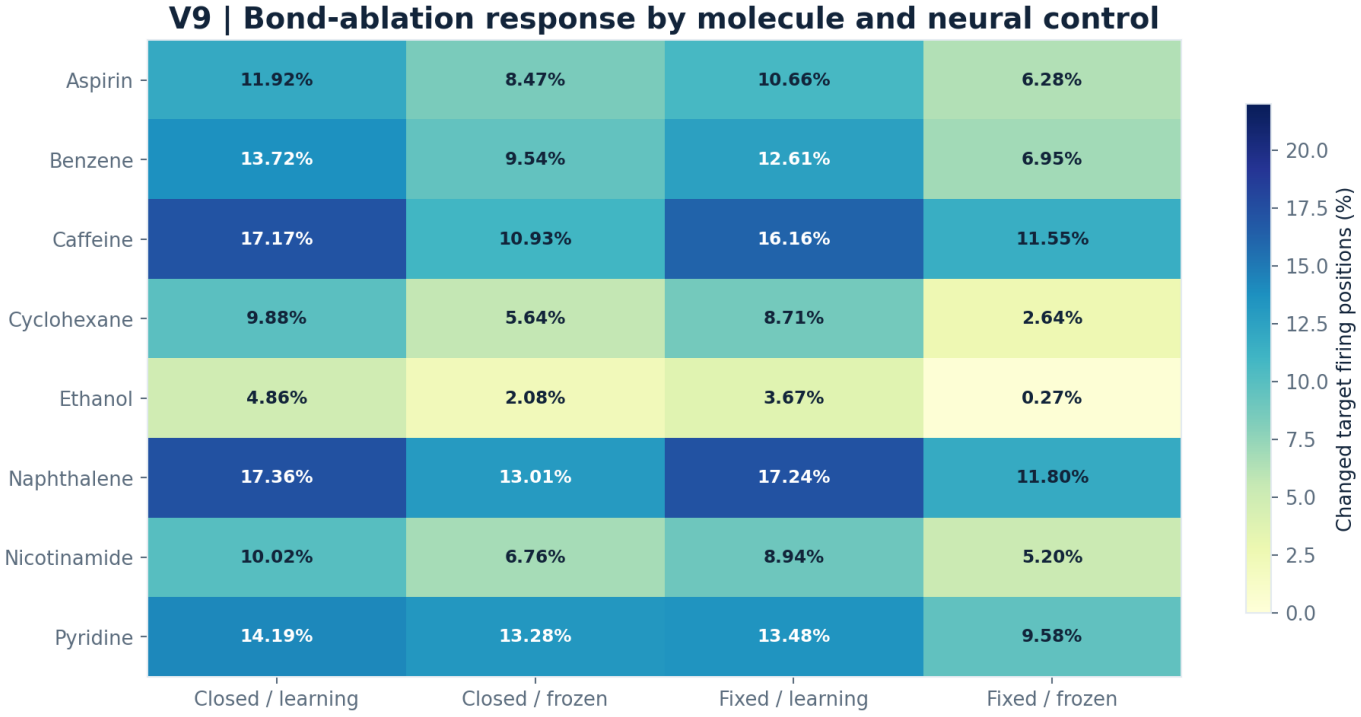


Source: GRAPH_CLASSIFICATION_CONDITIONS.csv

In V7–V8, the four newly added RNA cases did not show a statistically reliable graph-label decoder after the four-comparison multiplicity correction. These negative findings are reported, not discarded.

EXHIBIT 04 / SOURCE-RENDERED DATA

Eight compounds × four controls

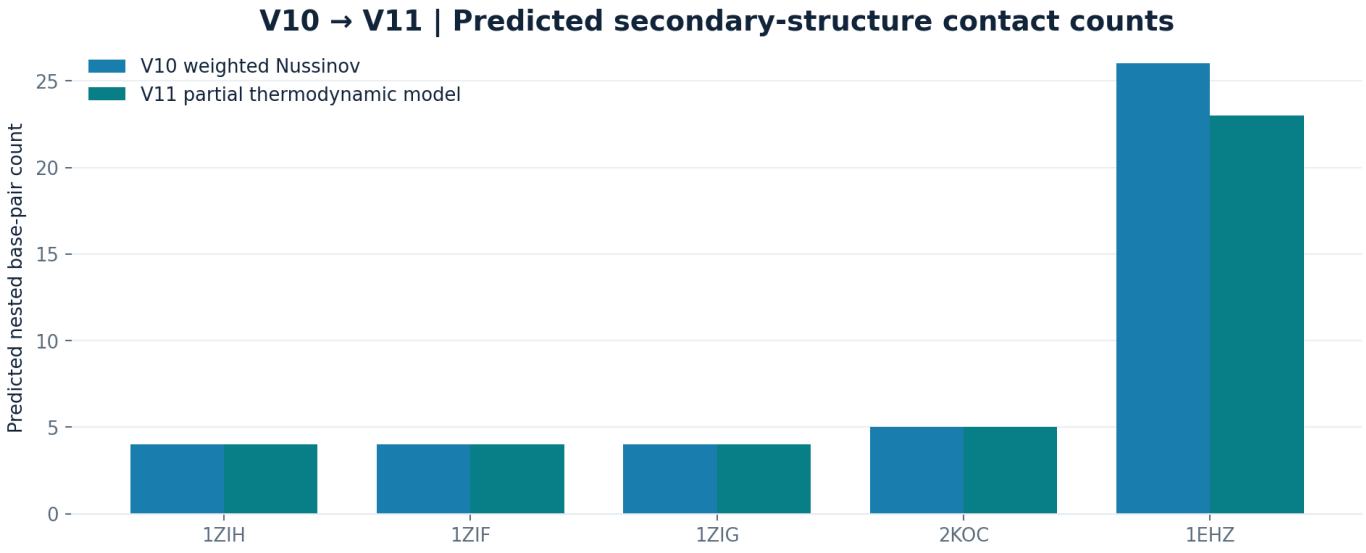


Source: ALL_CONDITIONS_FROM_RAW.csv; source original 64-seed ligand JSONL

V9 measures spike-event disagreement between intact and ablated chemical bond graphs. For example, caffeine was 17.17% under closed feedback and learning, not 17.17% chemical binding or predictive success.

EXHIBIT 05 / SOURCE-RENDERED DATA

RNA pair count changes across solver versions

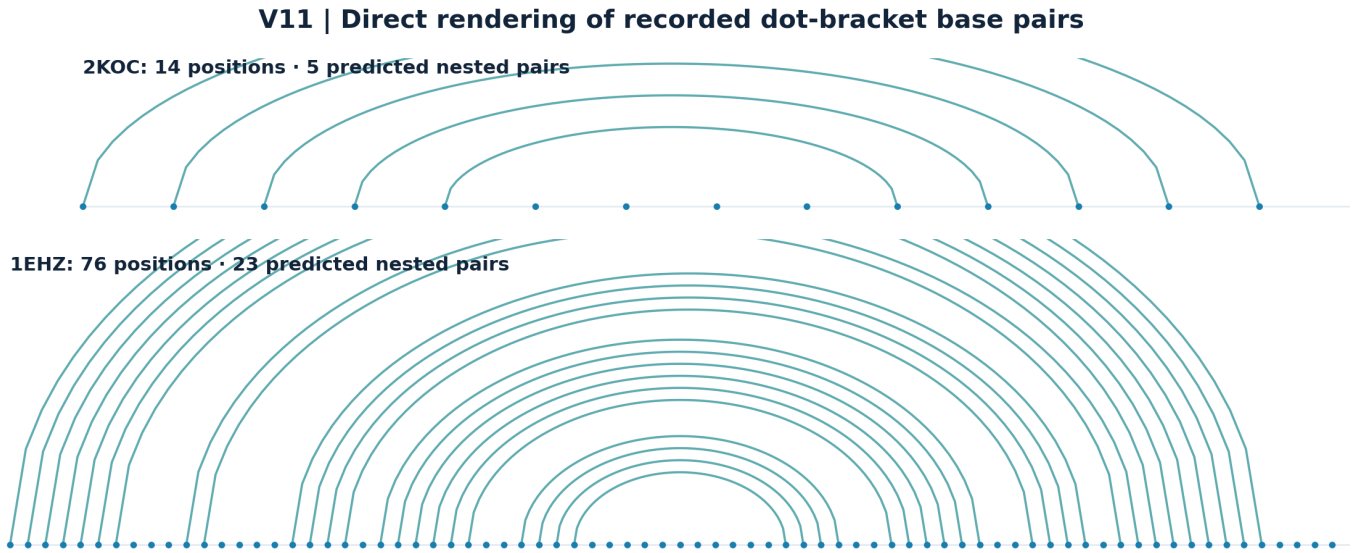


Source: RNA_FOLDING_STRUCTURES.csv; original calculated dot-bracket structures

V10 implemented a weighted noncrossing Nussinov solver. V11 used selected Turner-source nearest-neighbor energy terms and produced 40 nested pairs in five RNAs, versus 43 in V10. Neither pair count is an experimental accuracy score.

EXHIBIT 06 / SOURCE-RENDERED DATA

Base-pair arcs rendered directly from RNA output

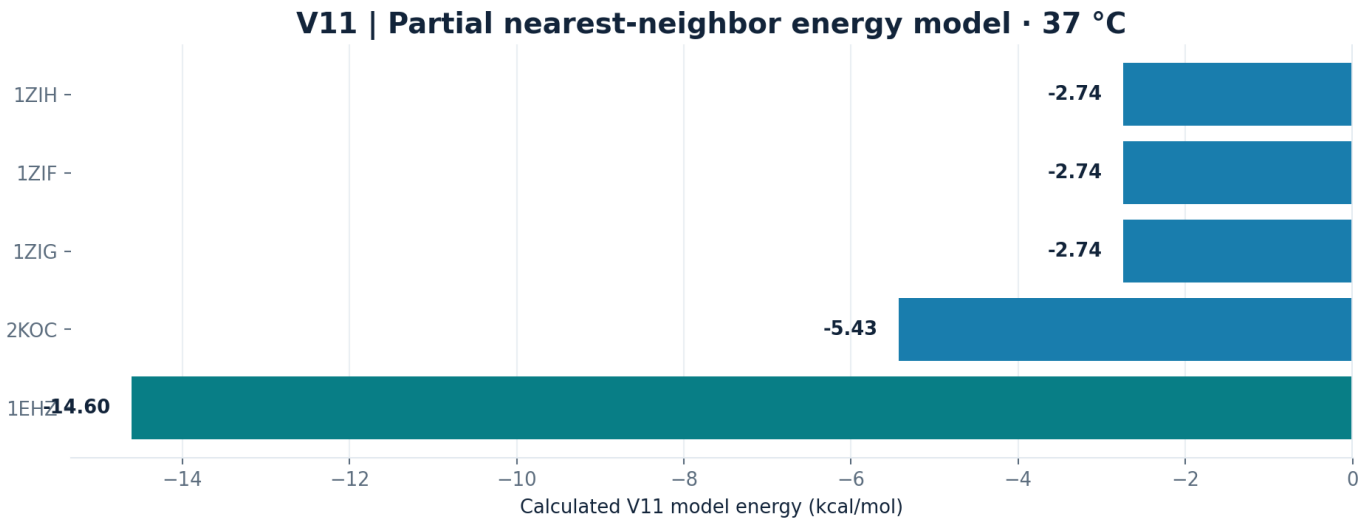


Source: RNA_FOLDING_STRUCTURES.csv; V11 dot-bracket fields

Every plotted arc is a paired opening/closing bracket from the saved solver output. Arc positions show RNA secondary structure connectivity, not actual atom coordinates or three-dimensional molecular dynamics.

EXHIBIT 07 / SOURCE-RENDERED DATA

Partial nearest-neighbor energies at 37 °C

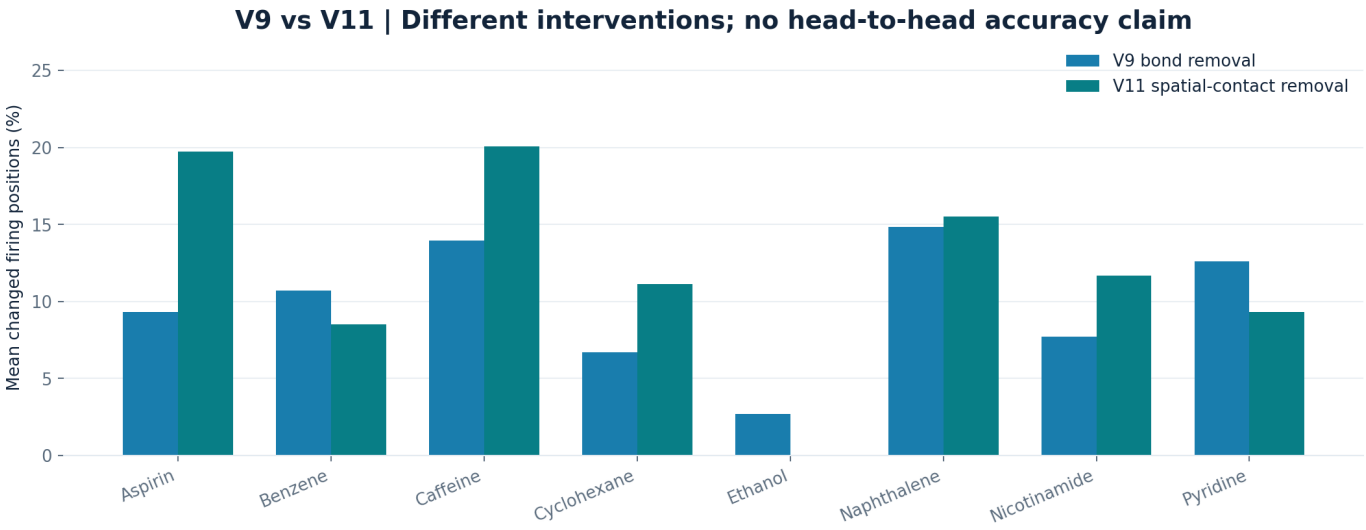


Source: RNA_FOLDING_STRUCTURES.csv

These five free-energy-like values are internal results of the coded partial thermodynamic model. They must not be compared with full RNAfold partition-function or full Turner thermodynamic outputs as equivalent physical folding energies.

EXHIBIT 08 / SOURCE-RENDERED DATA

Ligand response: different representations, different tasks

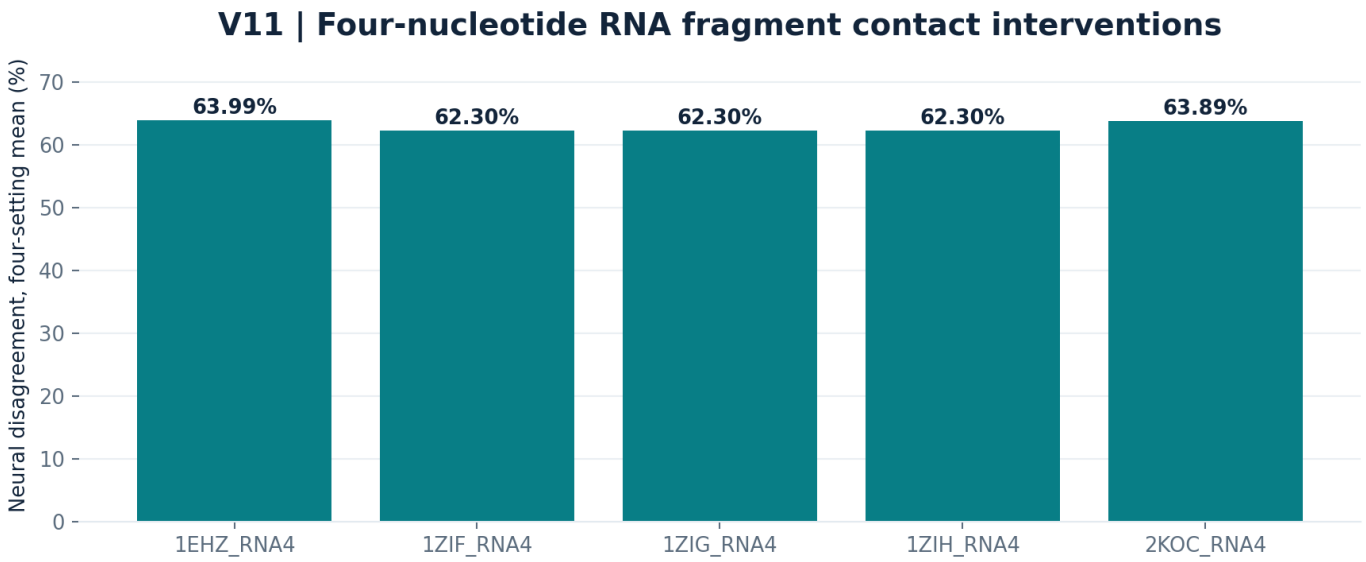


Source: ALL_CONDITIONS_FROM_RAW.csv

V9 removes chemical bonds while V11 removes ETKDG-derived spatial contacts. The graph highlights measured response profiles, not performance improvement. No ligand assays or binding labels appear in the native source evidence.

EXHIBIT 09 / SOURCE-RENDERED DATA

High response in short atomistic RNA fragments

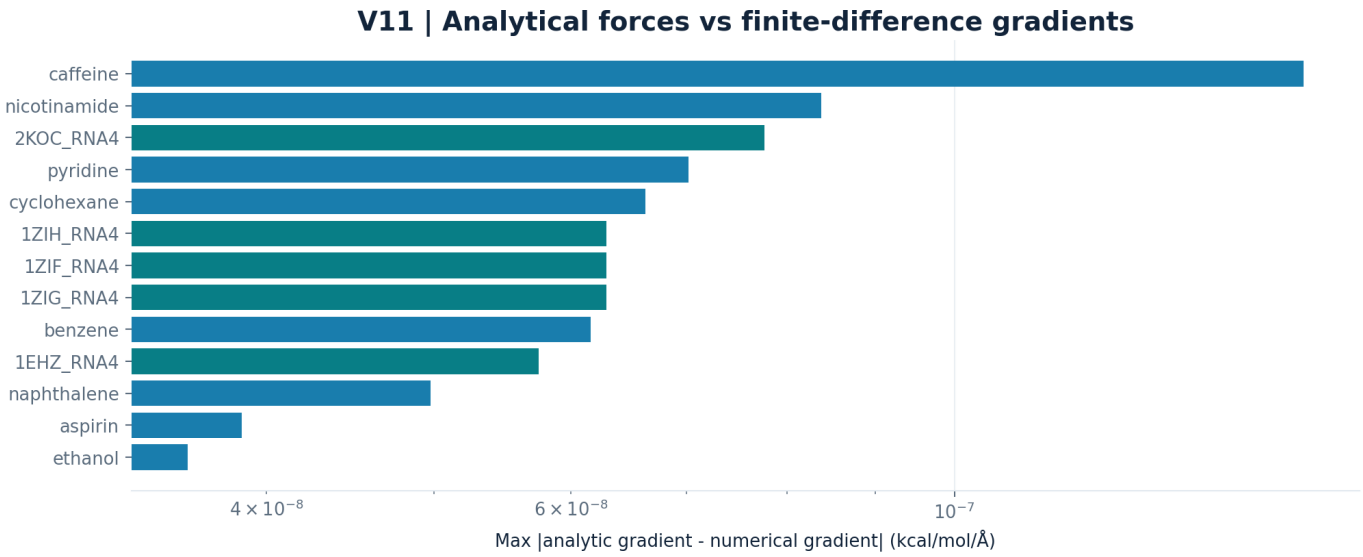


Source: ALL_CONDITIONS_FROM_RAW.csv

V11 performs MMFF94 on four-nucleotide RNA fragments, then removes hundreds of generated spatial contacts in a counterfactual. Approximately 62–64% neuronal disagreement does not mean 62–64% correct RNA folding.

EXHIBIT 10 / SOURCE-RENDERED DATA

Numerical force consistency: 13 molecules/fragments

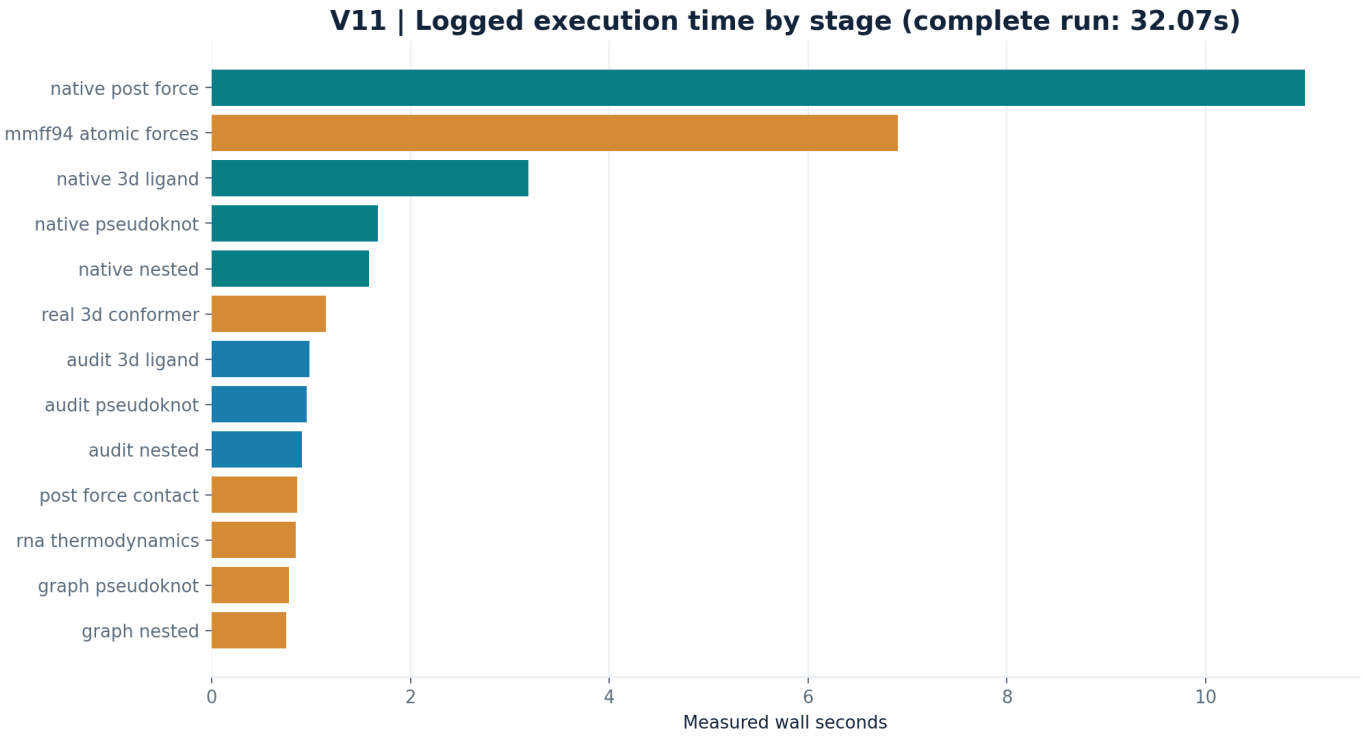


Source: ATOMIC_FORCE_RESULTS.csv

The recorded 13 minimizations converged. Sampled analytical MMFF94 gradients were compared to finite differences; the largest absolute discrepancy was 1.59×10^{-7} kcal/mol/Å. This validates numerical consistency, not a full biomolecular model.

EXHIBIT 11 / SOURCE-RENDERED DATA

Timing: the stages recorded in the machine log

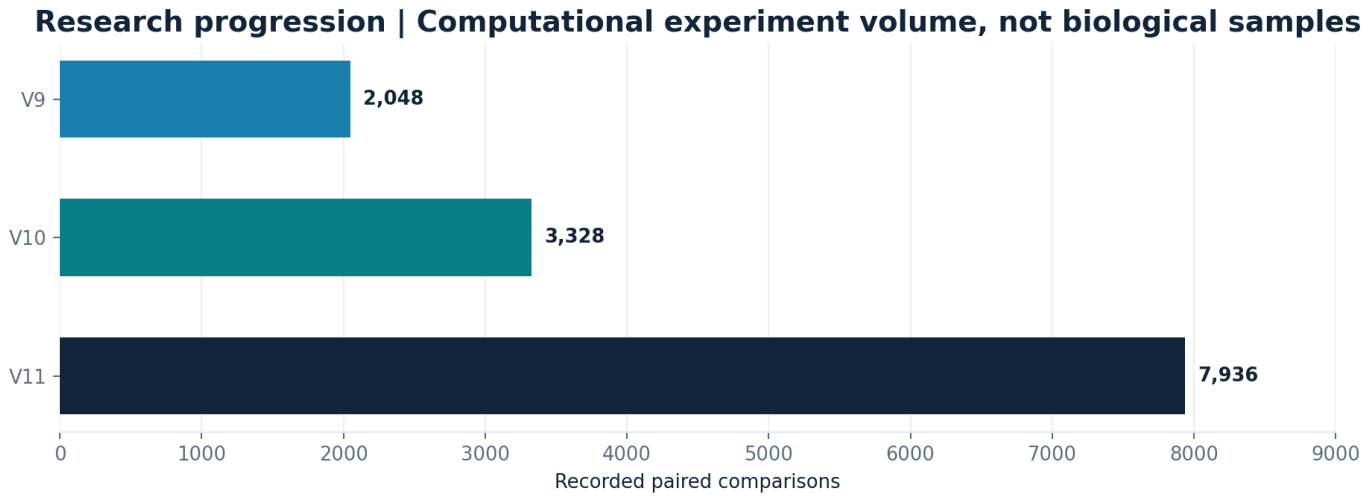


Source: RECORDED_TIMING_SECONDS.csv

The accepted 64-seed V11 pipeline took 32.067 seconds on the archived execution host. This is not an external speed comparison with ViennaRNA, BPfold, molecular dynamics or AlphaFold.

EXHIBIT 12 / SOURCE-RENDERED DATA

Computational workload grows, not biological samples

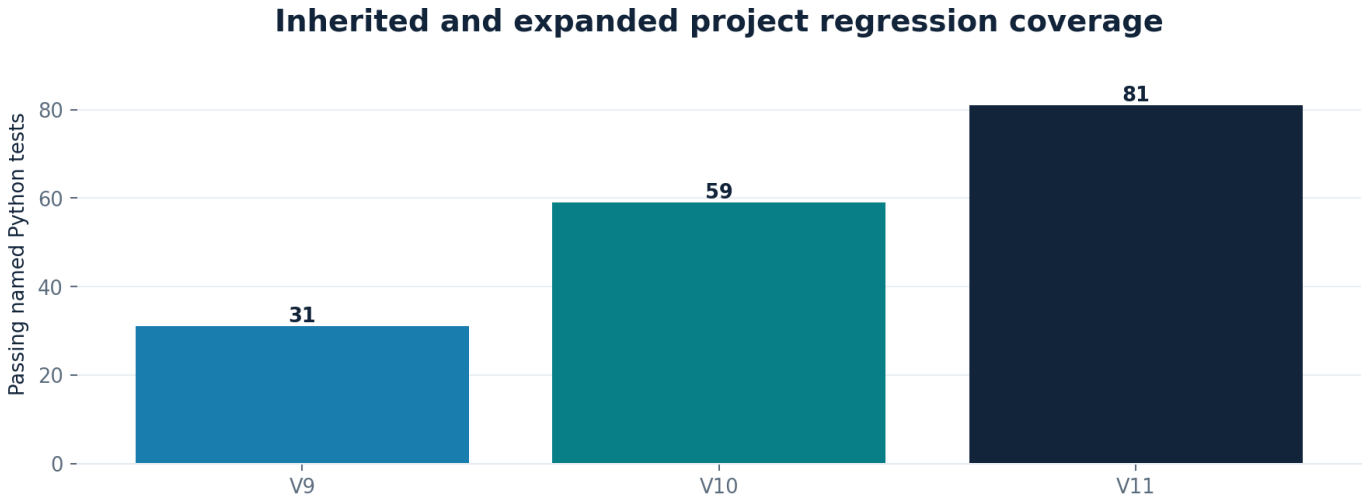


Source: ALL_V9_V10_V11_PAIRED_COMPARISONS.csv

The record counts are deterministic paired neural runs: V9 2,048, V10 3,328, V11 7,936. Their experimental questions differ, so no blanket aggregate accuracy is computed.

EXHIBIT 13 / SOURCE-RENDERED DATA

Recorded regression coverage



Source: TEST_CASE_INDEX.csv; original version-specific regression logs

V11 records 81 passing named project Python tests. Source receipts show a full V10 upstream pass; a later V11 one-shot upstream run was interrupted and followed by separate per-command checks. The latter is not relabeled a single complete pass.

The measurements behind every graph

Spike disagreement: D = Σ(t=0..47) popcount(A(t) XOR B(t)) / (48 × 22). The A and B arrays encode target-neuron spike masks for matched intact/ablated conditions. This quantity measures neural response differences, not folding accuracy.

Binary graph classifier: accuracy = correct predictions / evaluated graph labels. For balanced labels, the chance baseline is 50%; each perturbation and readout must be assessed with its own controls.

Permutation tail probability: p = (1 + number of null scores ≥ observed score) / (4096 + 1), followed by Holm correction for multiple tests. Null assumptions pertain to deterministic simulation seeds, not biological patient populations.

RNA model energy: G_model = selected stacking + loop initiation + simplified loop terms; only the coded subset of a mature Turner/ViennaRNA model is implemented. A selected H-type pseudoknot candidate is not experimentally confirmed.

Force-gradient check: compare analytical Cartesian gradient ∂E/∂x with [E(x+h) − E(x−h)]/(2h), as sampled and recorded. V11 used RDKit MMFF94 on 8 small molecules and the first 4 residues of each of 5 RNAs.

Checksums: SHA-256(original data file) versus the archived expected digest protects against accidental source replacement. Agreement shows matching bytes, not an independent molecular assay.

Evidence and reproducibility instructions

See data/FIGURE_PROVENANCE.csv for each PNG, its source CSV, transformation and SHA-256. data/ contains the individual paired runs and derived condition tables. evidence_and_logs/ contains original test/build logs plus the unchanged source-audit evidence ZIP. Run the original verifier in that evidence archive to recompute firing-mask parity. The archived executable source tree can rerun native computations with its prescribed toolchain.

State of the field—and exactly what is comparable

METHOD / STUDY	EXTERNALLY REPORTED RESULT	COMPARABILITY
BPfold (2025)	F1 0.820 on ArchiveII and 0.658 on bpRNA-TS0	Experimentally annotated RNA base-pair prediction; N has no corresponding F1 benchmark
G4ShapePredictor (2024)	76% ± 1% accuracy for three G4 folding classes	Not comparable with N 76.56% for constructed graph intervention
ViennaRNA RNAfold	MFE structure, full calibrated thermodynamic model and partition functions	N V11 implements selected terms only
RNAstructure / NNDB	Turner 2004 thermodynamic parameter families	Parameter provenance; not proof of complete implementation
RDKit ETKDG/MMFF94	Chemical conformer embedding and molecular mechanics	N actually calls these established methods for 3D computations
RCSB PDB 1EHZ & 1KF1	Experimental solved structures; X-ray coordinate records	Molecular identity provenance, not a held-out structure prediction test

Published accuracy values are labeled as external literature results, not N measurements. Comparisons require the same tasks, held-out ground truth, input conditions, metrics, and computing environment.

Primary references

Rochester NNDB / Turner 2004 — https://rna.urmc.rochester.edu/NNDB/

RNAstructure parameters — https://rna.urmc.rochester.edu/Overview/Parameters.html

ViennaRNA RNAfold — https://www.tbi.univie.ac.at/RNA/ViennaRNA/doc/html/usage/man/RNAfold.html

BPfold, Nature Communications 2025 — https://www.nature.com/articles/s41467-025-60048-1

G4ShapePredictor, Scientific Reports 2024 — https://www.nature.com/articles/s41598-024-74826-2

RDKit — https://www.rdkit.org/docs/

RCSB PDB 1KF1 — https://www.rcsb.org/structure/1KF1

RCSB PDB 1EHZ — https://www.rcsb.org/structure/1EHZ

Demonstrated computation. Open scientific validation.

VERIFIED / RECORDED

• Compiling and executing the retained native spiking neural engine with structure-derived contact graphs.

• 13,312 paired response records reconstruct from original masks with zero disagreements.

• V6 structural graph contrasts include positive and stringent negative controls.

• RNA secondary-structure calculation, an H-type pseudoknot candidate search, RDKit geometry optimization, and MMFF94 numerical force checks execute in source.

• 81 named V11 project tests have passing historical log entries.

• Independent replay and source hashes support deterministic computational reproducibility.

NOT SHOWN BY THESE RUNS

• Blind experimental secondary-structure base-pair precision/recall or F1.

• General whole-RNA 3D atomic folding accuracy, RMSD, solvent/ion-calibrated dynamics or drug–DNA binding affinity.

• Fair same-task performance superiority against ViennaRNA, RNAstructure, BPfold, G4ShapePredictor or molecular dynamics.

• Original project Git history: the available .git recovery is a newly constructed offline snapshot.

This portfolio is a transparent computational research artifact. It is deliberately not presented as peer-reviewed proof of biological prediction.